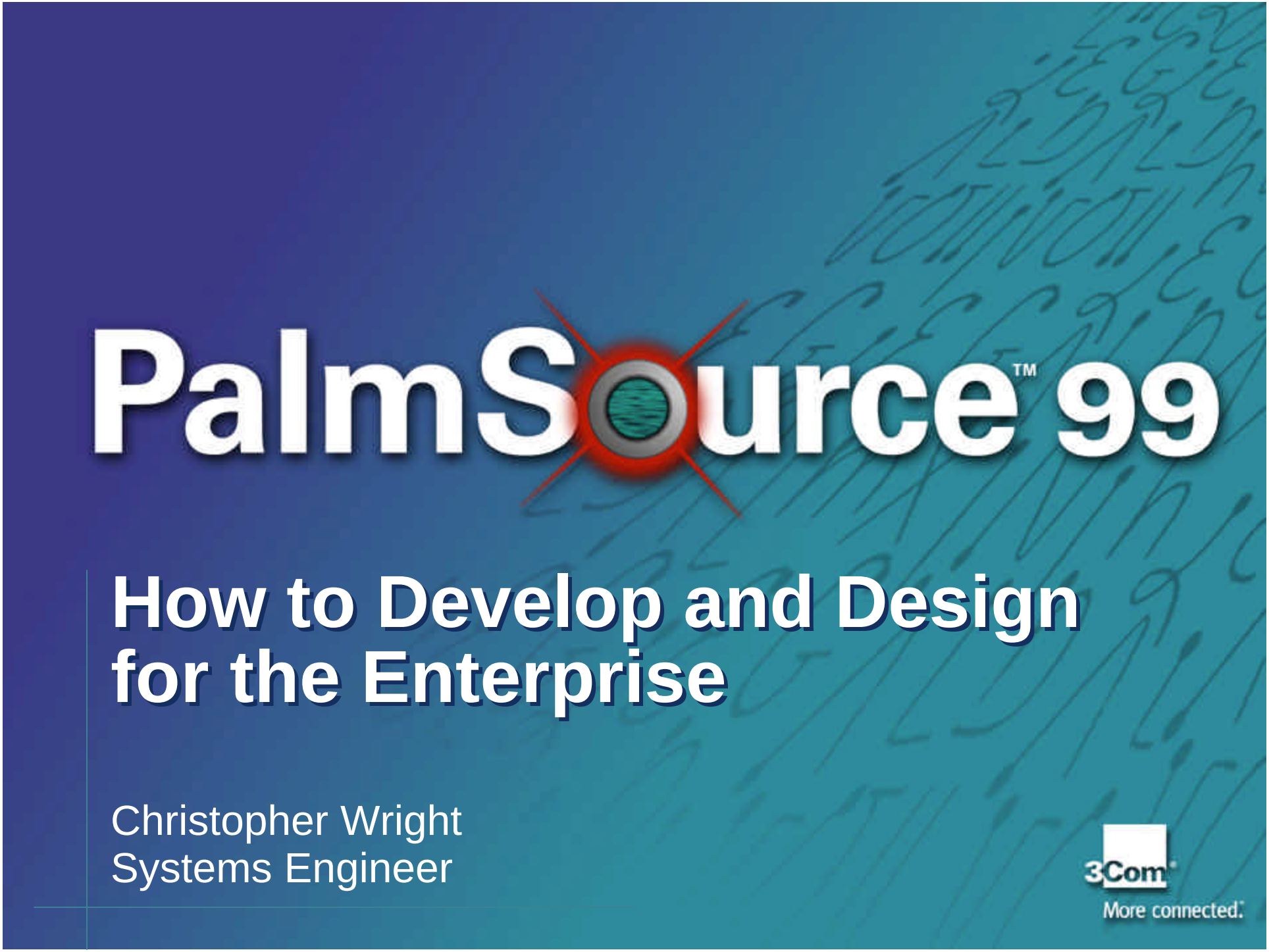
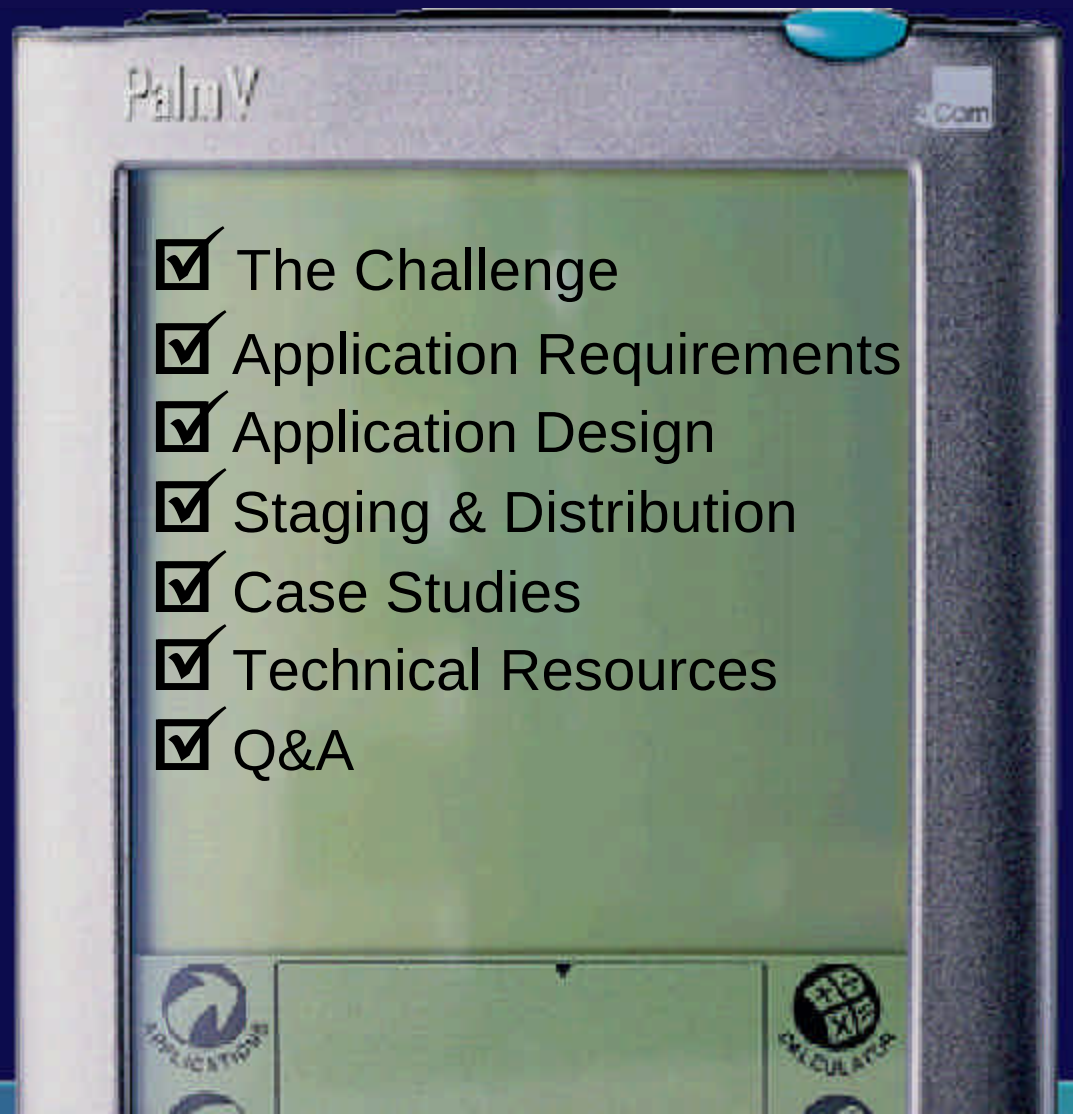

PalmSource™ 99
How to Develop and Design for the Enterprise
Christopher Wright
Systems Engineer
3Com
More connected.

Agenda

PalmSource 99



- ☑ The Challenge
- ☑ Application Requirements
- ☑ Application Design
- ☑ Staging & Distribution
- ☑ Case Studies
- ☑ Technical Resources
- ☑ Q&A

The Challenge

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- 5000 users -----> 10,000
- Immediate access to corporate data
- Must work with existing systems
- Encrypted, password protected
- Wireless / Wireline
- Fast, efficient, easy to use
- Need it tomorrow

The Pieces Are in Place

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- Middleware exists
  - Oracle
  - Sybase
  - DB2
  - SAP
  - Notes
  - Web
- Hardware choices
- The ROIs are large
  - Paper processes are inefficient
    - Printed data is old
    - Requires manual re-entry
    - Lost data

Application Requirements

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- How will the users synchronize?
- Partitioning an application
- Security
- Data entry options
- Hardware choices

Application Requirements
How Will the Users Synchronize?

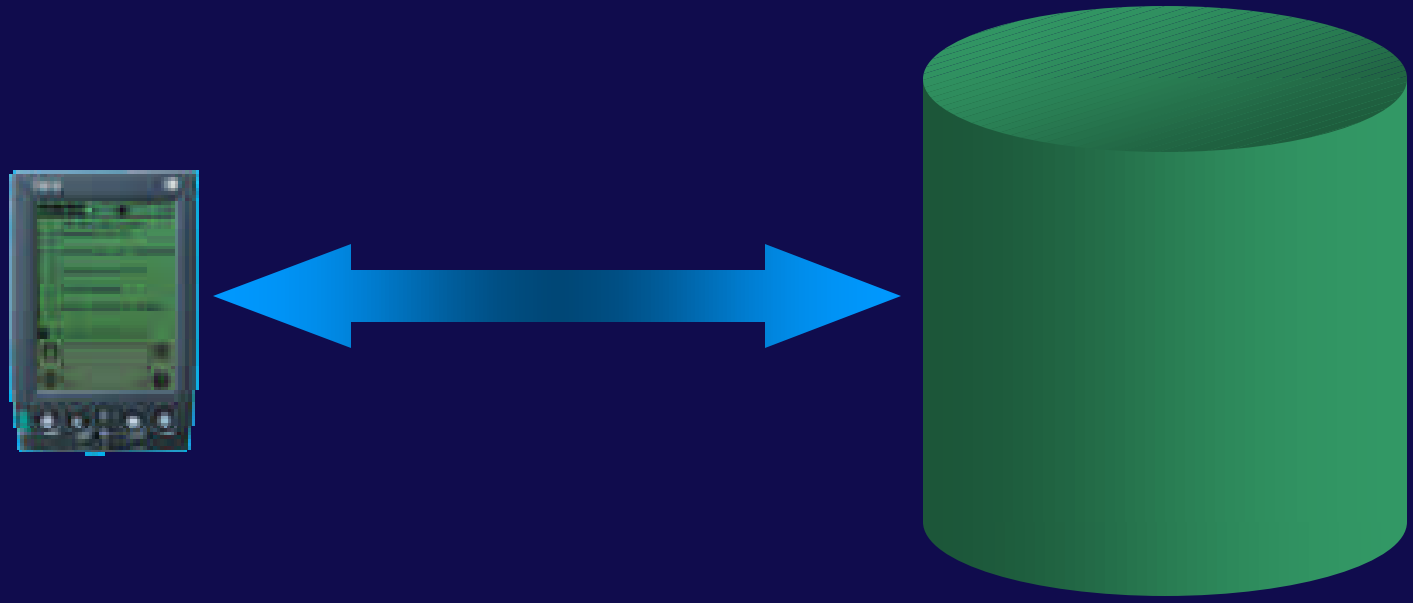
PalmSource 99

- Batch mode vs. real time
  - Can the data be old?
  - $$ and complexity
- Connectivity
  - Palm™ Cradle
  - Analog Modem
  - Wireless LAN
  - Wireless WAN
- Mobile vs. Stationary

Application Requirements
Partitioning an Application

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- Local storage vs. server storage
- What portion synchronizes with Palm OS
  - User specific content



Application Requirements
Security

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- Device security
- Data encryption
  - Device databases
  - Over the air
- User authentication

- Medical records, stock transactions, Gov. data?

Application Requirements
Data Entry

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- Maximize check boxes and pick lists
- Speed data entry
  - Bar Code
  - Mag stripe
  - Infrared
  - Data Acquisition
  - Snap on Keyboard

Application Requirements
Hardware Peripherals

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- Rubber boot
- Analog / Wireless modem
- Infrared printer
- GPS
- Vehicle mounts
- Ethernet cradle / IR device

Application Design

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- 3 Development architectures
  - Forms based
  - C / C++
  - Web
- Middleware for enterprise systems
- Server synchronization
- 4 Network architectures
  - Local
  - Server synchronization
  - AvantGo Server synchronization
  - Palm VII / Palm.Net

Application Design
Forms Based

PalmSource 99

- Pros
  - Minimal conduit development
  - Customizable
  - Rapid development
  - Device side control / logic
- Cons
  - Run time fees
  - Some UI restrictions

Application Design
C / C++

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- Pros
  - Efficient / fast application
  - Large pool of experienced developers
    - >20,000 registered with Codewarrior
  - No run time fees
- Cons
  - Conduit development required
  - Experienced programmer required

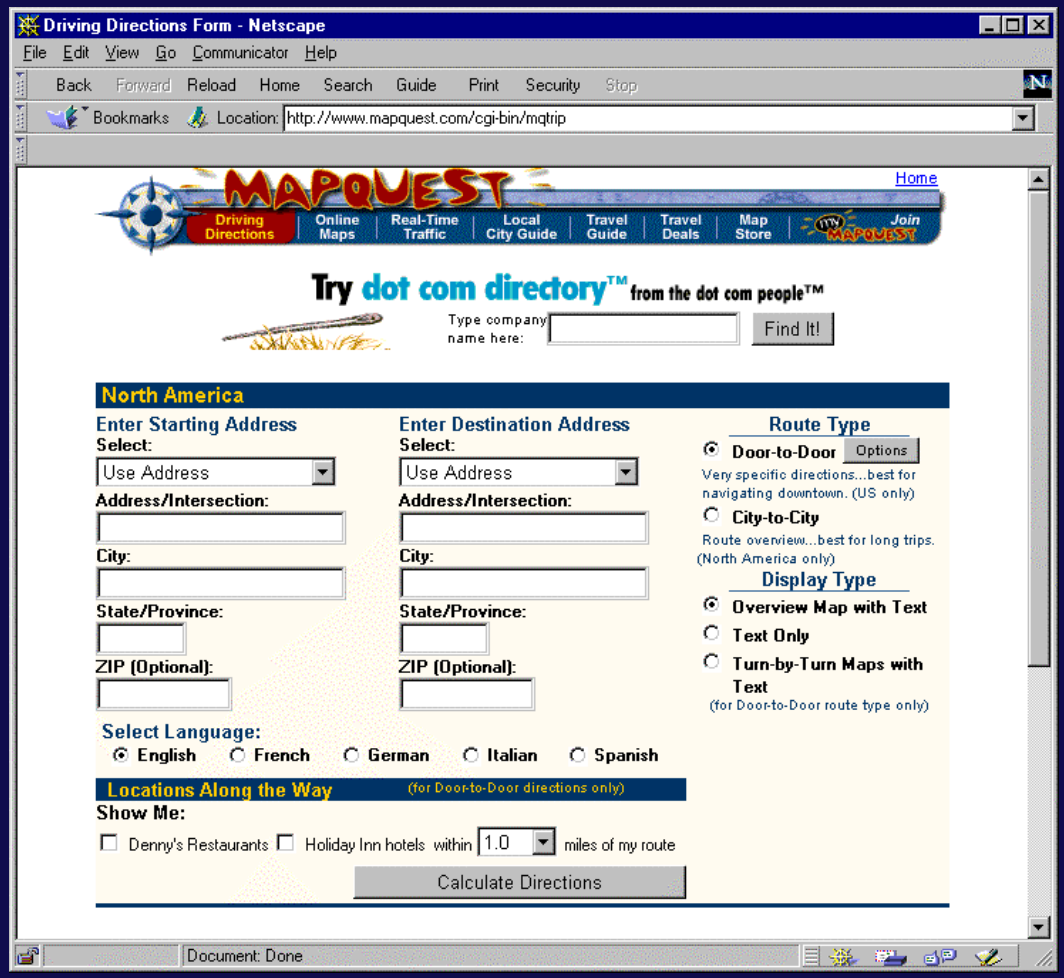
Application Design
Web

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- Pros
  - No conduit development required
  - Development in HTML
  - Use existing web pages
  - Supports batch and real-time modes
  - Easily distribute application changes
- Cons
  - Limited device side logic

Application Design
Make the Web Palm Friendly

PalmSource 99





Application Design
Solutions Exist for Enterprise Data

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- Oracle
  - Oracle Lite
- Sybase
  - Sybase Ultralite
- DB2
  - DB2 Everywhere
- Legacy (OS400, Terminal emulator)
  - Attachmate, Translink
- Web
  - AvantGo, Palm.Net™ service

Application Design
Server Synchronization Tools

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- AvantGo
- Synchrologic
- Riverbed Scout / Palm HotSync Server
- WaveWare
- Intellisync Anywhere
- Mobile Connect / ASL-Connect
- Pylon ServerSync

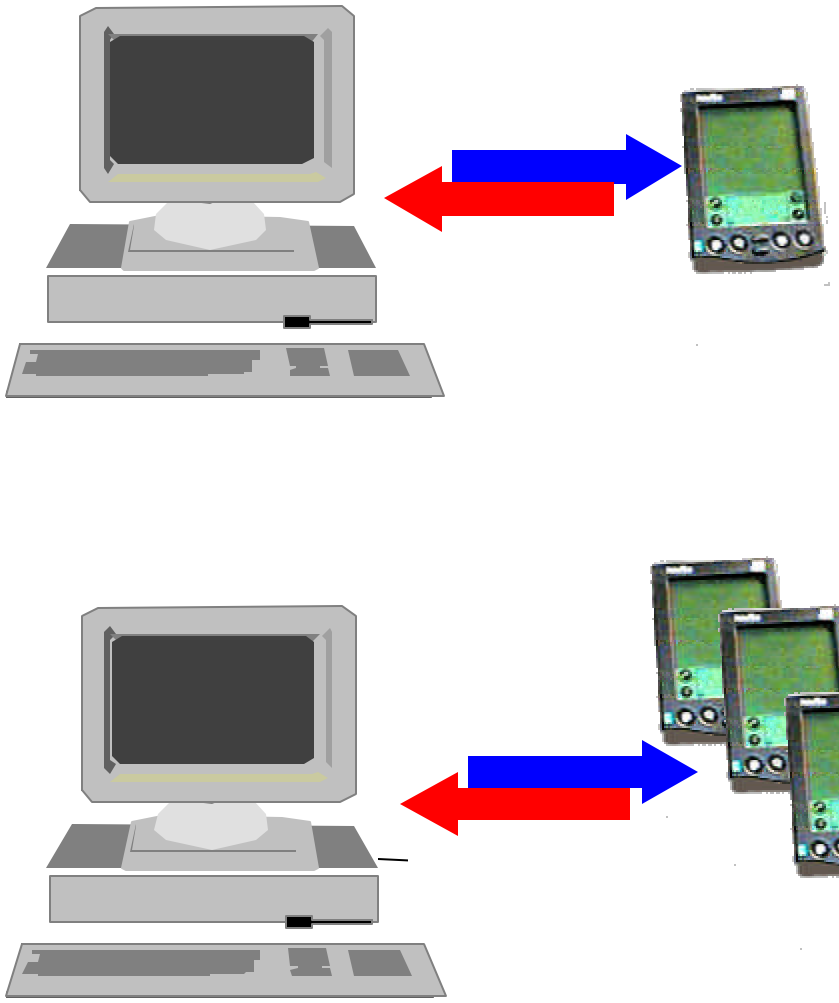
Application Design
4 Network Architectures

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- Local PC synchronization
- Server synchronization
- AvantGo Server synchronization
- Palm VII™ device/ Palm.Net™ service

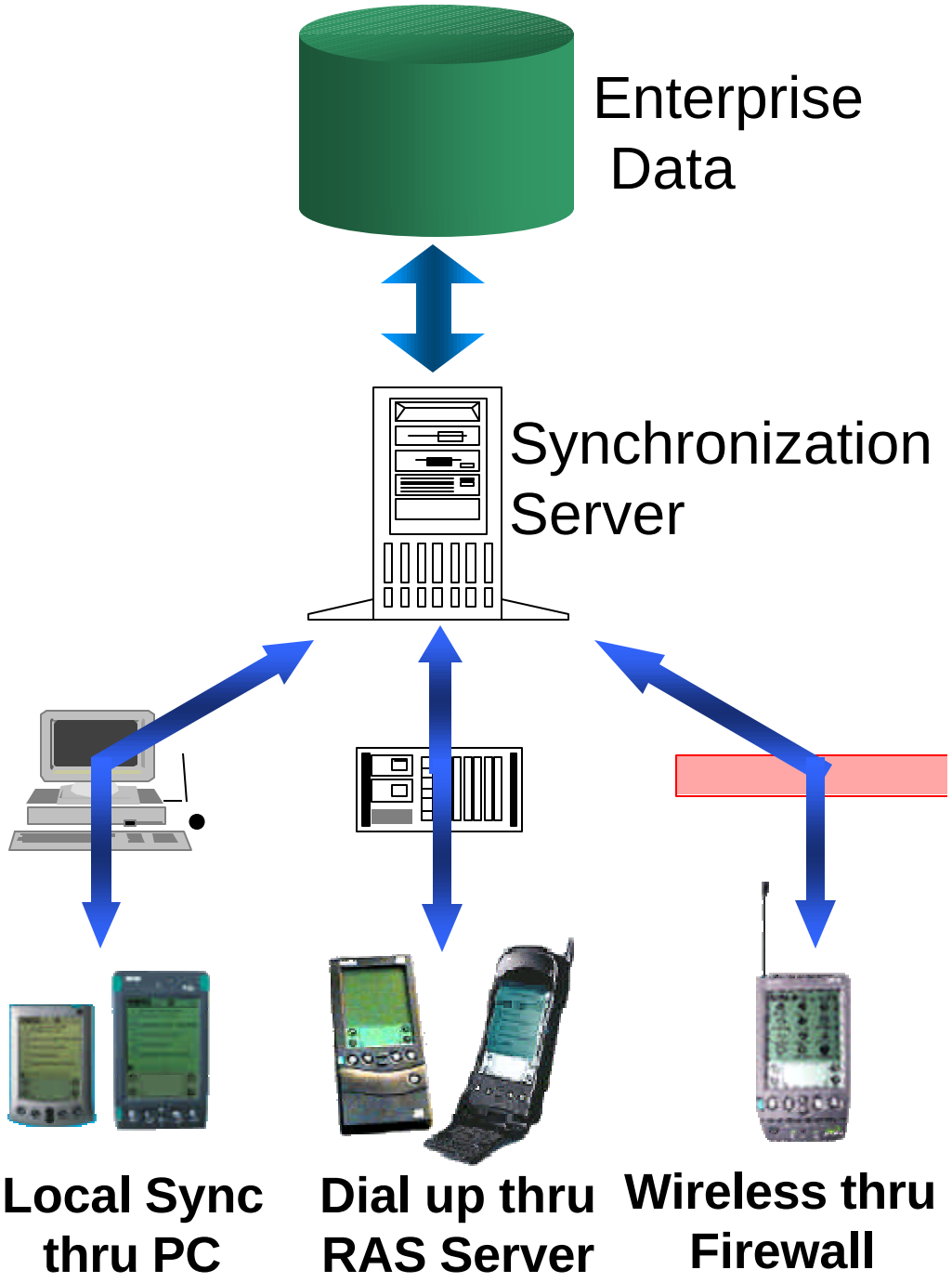
Application Design
Network Architecture #1

- Local PC Synchronization
  - One synchronization at a time
  - Small PC application
  - Batch mode
  - Non-wireless



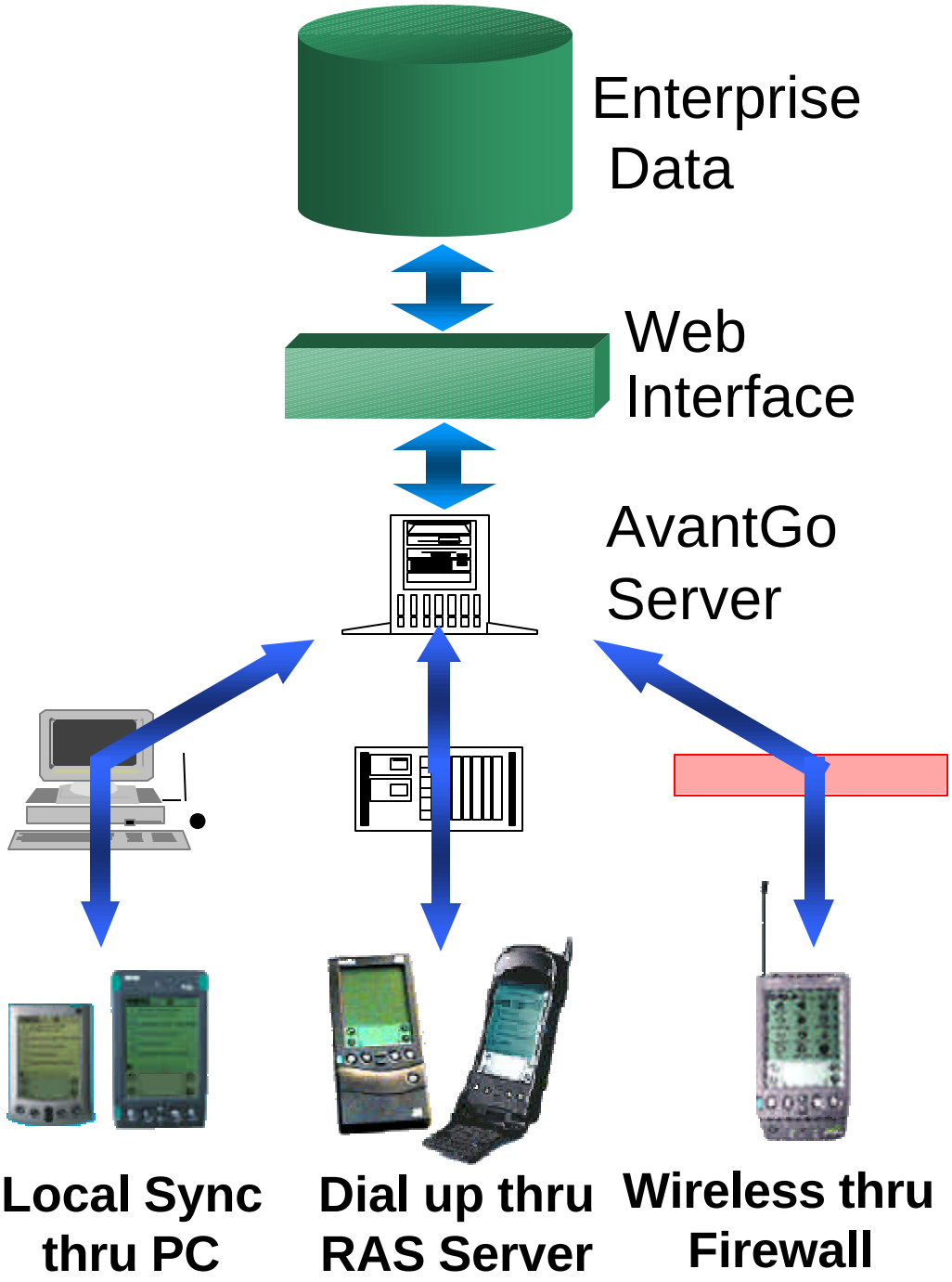
Application Design
Network Architecture #2

- Server Synchronization
  - Simultaneous Synchronization
  - Scalable
    - 1000s of users
    - Multiple Palm OS devices
  - Networked Environment
  - Secure
  - Batch mode



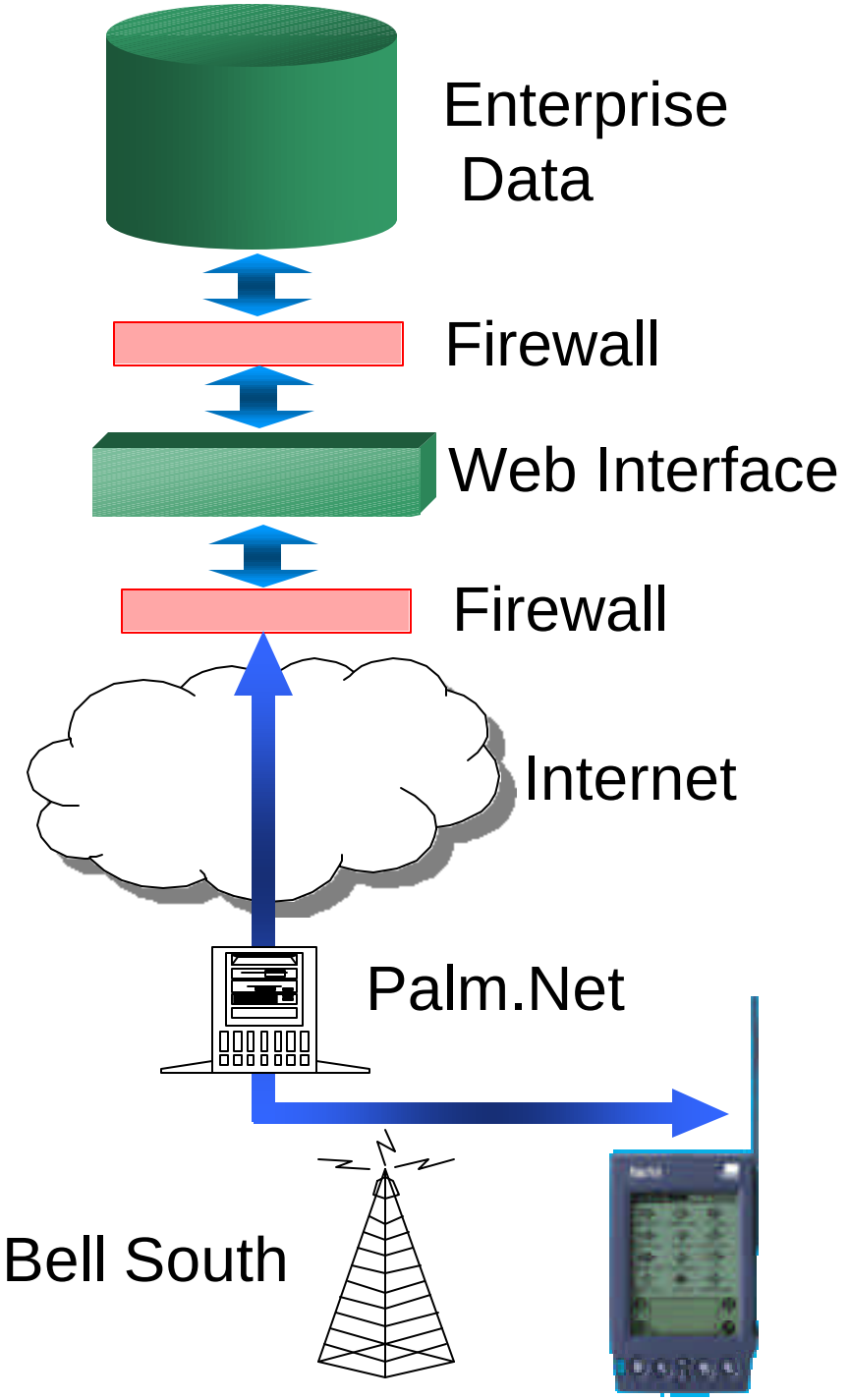
Application Design
Network Architecture #3

- AvantGo Synchronization
  - Leverage Intranet investment
  - Batch or real time



Application Design
Network Architecture #4

- Palm VII™ device/ Palm.Net™ service
  - Real Time
  - Secure
  - Minimal Development



Application Distribution and Staging

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- Building a device image
- Application distribution

Application Distribution and Staging
Building a Device Image

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- How to Image 5000 devices?
  - 1) Build the perfect Palm
  - 2) Backup Buddy—obtain every prc/pdb from the device
  - 3) Setup Sync stations, place prcs/pdbs in backup folder
  - 4) Sync til you drop
  - 5) Remove the username?

- With applications in Flash
  - TRG

Application Distribution and Staging
Application Distribution

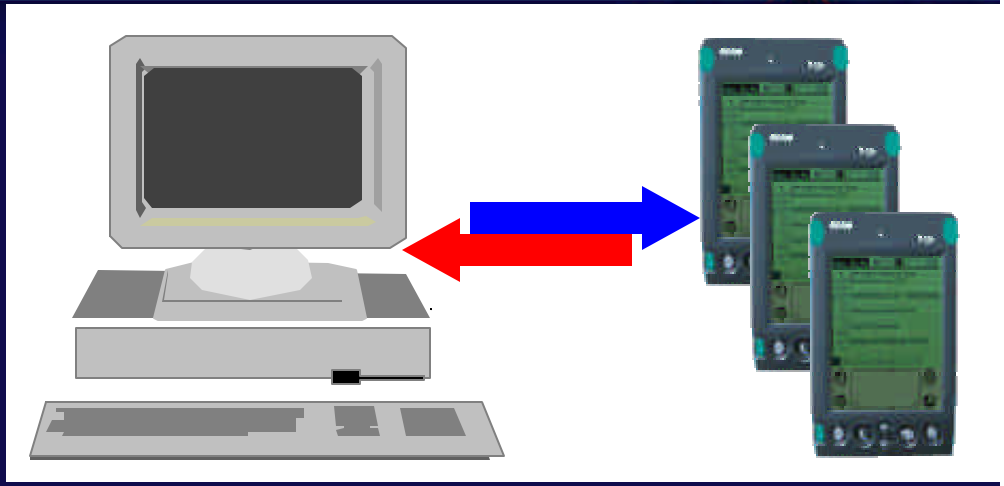
PalmSource 99

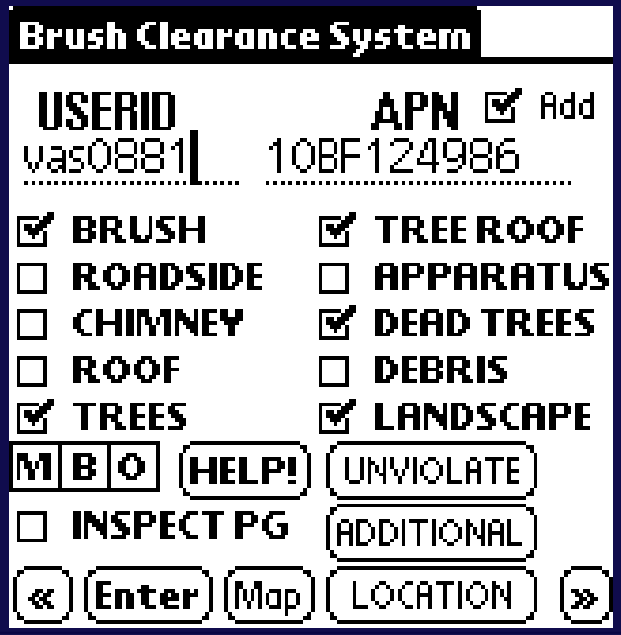
- Using CA
  - shipIT
  - aimIT
- Using Server synchronization
- E-mail new prcs??

Case Study #1
LA City Fire Inspection

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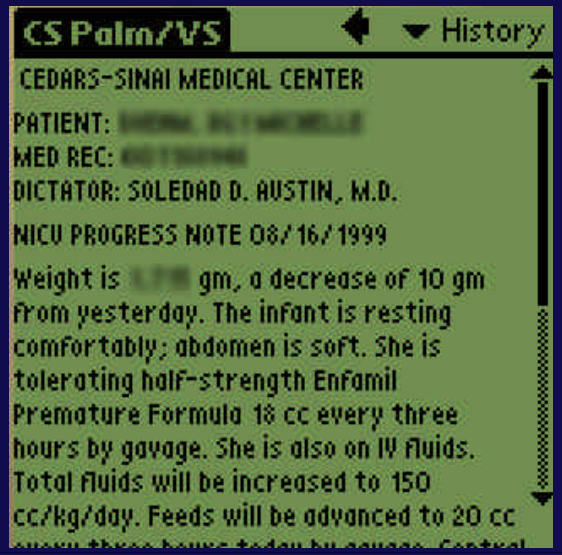
- Forms-based application
- Local Synchronization
- Efficient data entry
- Batch mode
- Specifics
  - MS Access Database
  - Symbol 1500
  - Satellite Forms
- ROI

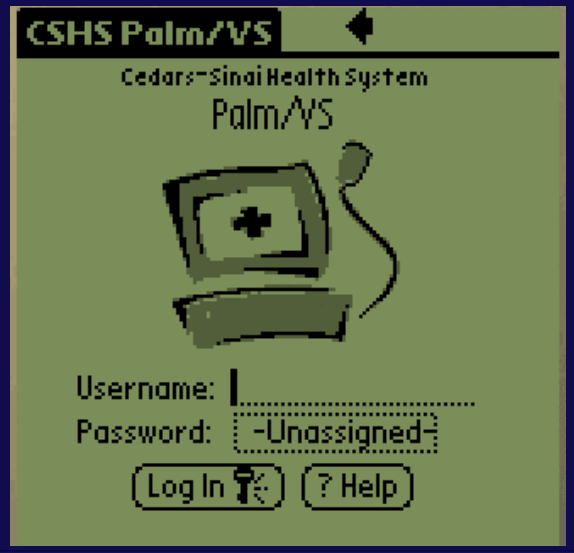


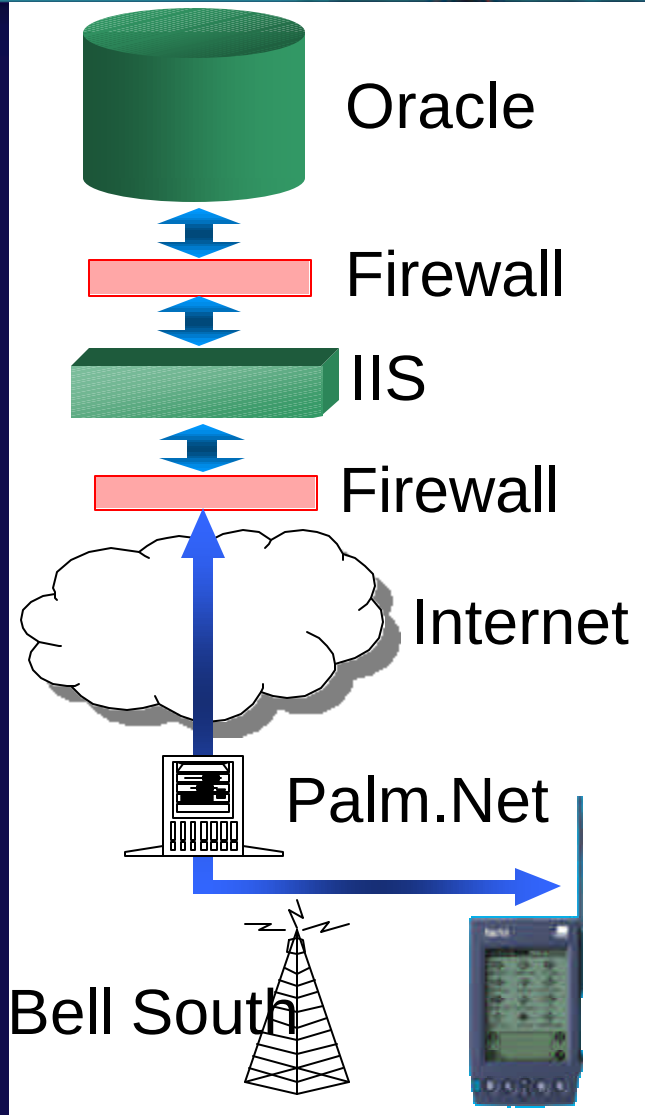


Case Study #2
Cedars-Sinai Palm/VS

- Doctors can query
  - Transcribed reports
  - Lab data
- Secure
- Real time

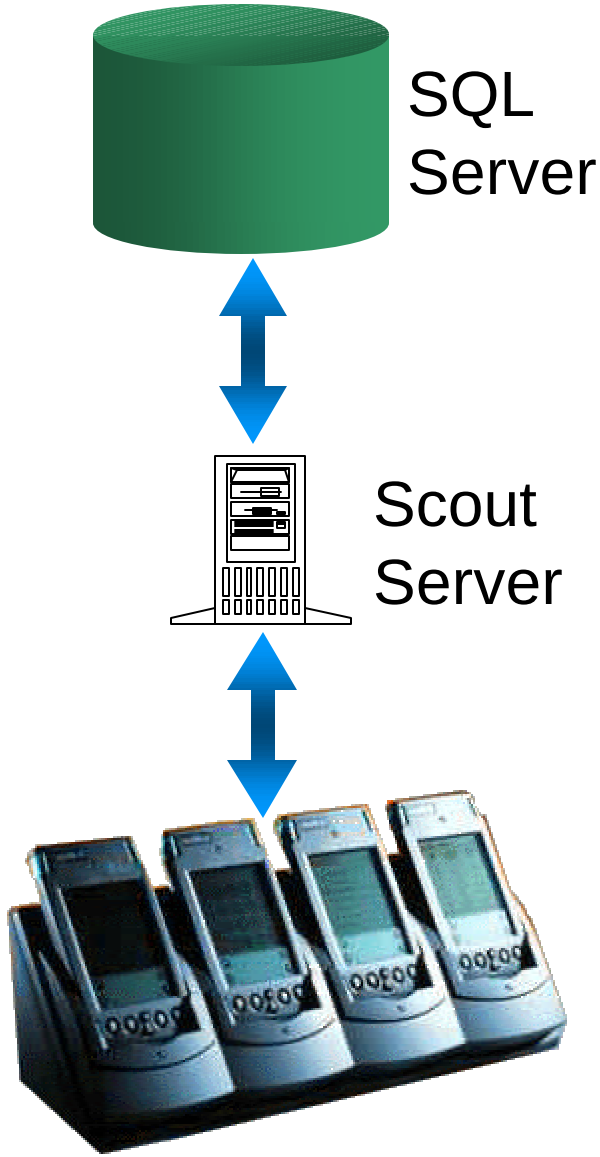






Case Study #3
US Postal Office

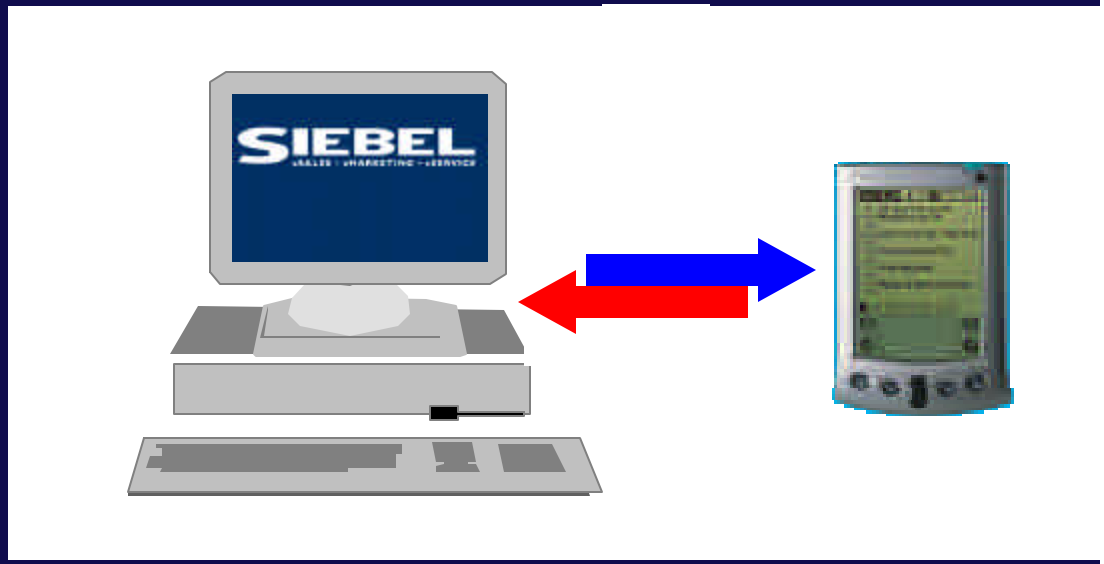
- CodeWarrior C / C++ application
- 12 airports each with a Symbol Ethernet cradle
- 1 Scout Server
- Scalable



Case Study #4
Cisco Systems— SFA

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- Metrowerks C / C++
- Used existing address book source code
  - Windward with Synchrologic



Technical Resources

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- Palm Computing resources
- Third-party resources


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More connected.